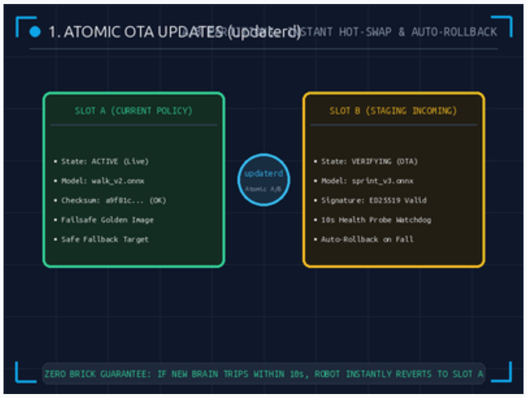
■ MICRODUCK PHYSICAL AI MASTERCLASS

PHASE 6 HANDOUT

Module 6: Securing the Swarm: OTA Updates (updaterd), MuJoCo CI Gates, Cryptography & Fleet Observability

DevSecOps

UPDATES: A/B Atomic (updaterd)	CI GATE: 1,000 MuJoCo Sims	CRYPTO: ED25519 + SHA-256	FLEET: Swarm Observability



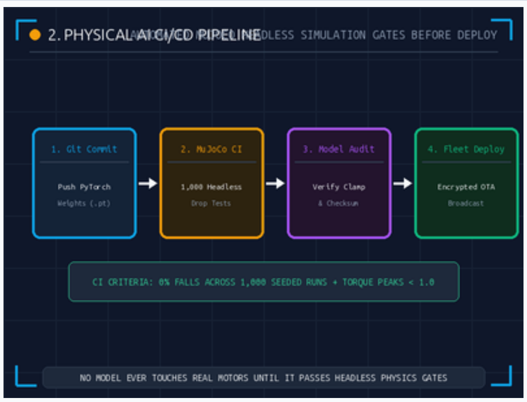
■ 1. Atomic OTA Updates & Rollback

A/B PARTITIONING & ZERO-BRICK FAILSAFE

Over-The-Air Brain Swaps: `updaterd` downloads candidate policies to Slot B while Slot A runs. A 10-second post-update health watchdog automatically rolls back to Slot A if the robot trips.

The 12-Year-Old Analogy: Neo plugging in a Kung Fu data cartridge in The Matrix. If the move has a glitch, the duck immediately snaps back to its safe old reflexes.

- **A/B Partitioning**: Guarantees atomic, seamless hot-reloading without interrupting ongoing operations.
- **10s Health Watchdog**: Monitors IMU angular acceleration; trips trigger instantaneous fallback to golden slot.
- **Zero-Brick Guarantee**: Hardware never gets stuck with a corrupted or non-functional policy file.



■ 2. Physical AI CI/CD Testing Gates

1,000 HEADLESS MUJOCO SIMULATION RUNS

Automated Physics Gating: On every git commit, GitHub Actions executes 1,000 parallel headless MuJoCo drop tests across friction, mass, and slope variances to verify 100% stability before deployment.

The 12-Year-Old Analogy: A digital gymnastics tryout. The new brain must balance through 1,000 simulated obstacle courses without falling once before touching real motors.

- **Physics Test Gate**: Runs 1,000 headless physics seeds on GPU cluster in <15 seconds.
- **Torque Spike Audit**: Asserts that peak motor commands never exceed normalized $[-1.0, 1.0]$ bounds.
- **Automated Rejection**: Any simulated fall immediately blocks model signing and halts the release pipeline.

■ 3. Edge Security & Anti-Tamper

ED25519 SIGNATURES & ROOT OF TRUST

Cryptographic Policy Integrity: Edge robots verify SHA-256 checksums and ED25519 asymmetric signatures. Unsigned or modified ONNX binaries are rejected at the hardware root of trust.

The 12-Year-Old Analogy: A royal wax seal on an official letter. If a sneaky attacker changes even a single number in the brain, the seal breaks and the robot blocks it.

• **Asymmetric ED25519**: Public key on Rockchip RK3566 validates private signature generated during CI build.
• **Anti-Tamper Lock**: Protects physical robots from rogue WiFi man-in-the-middle model injections.
• **Immutable Root**: Read-only boot partitions prevent malware persistence across robot reboots.

■ 4. Fleet Observability & Diagnostics

FALL-RATES, LATENCY JITTER & THERMALS

Real-Time Swarm Health: Operating a fleet of 100+ robots requires streaming latency metrics, IMU tilt distributions, motor temperatures, and fall-rate anomalies to a cloud dashboard.

The 12-Year-Old Analogy: A fitness tracker smartwatch for your robot. If motor #4 gets too warm, the dashboard alerts the technician before the plastic melts.

• **Predictive Diagnostics**: Detects mechanical gear friction and battery degradation weeks before failure.
• **Swarm Fleet Sync**: Simultaneously updates entire fleets with synchronized policy rollout waves.
• **Zero-Trust Mesh**: mTLS encrypted MQTT/gRPC telemetry ensures fleet communication cannot be snooped.

■ CONGRATULATIONS! COMPLETE PHYSICAL AI MASTERCLASS GRADUATION

• **Full Stack Mastery:** Hardware (P1) $\to$ MuJoCo (P2) $\to$ PPO RL (P3) $\to$ ONNX Clamping (P4) $\to$ Rust Daemons (P5) $\to$ DevSecOps (P6).
• **Production Ready:** You have architected an enterprise-grade, secure, 50Hz real-time robotics deployment pipeline.
• **Next Step:** Connect to physical hardware via robotctl and watch the Microduck walk!

■ PHASE 6 KNOWLEDGE CHECK

ASSESSMENT

Test your understanding of DevSecOps, OTA Updates, Cryptography, and Fleet Observability.

5 Questions

Q1: Atomic A/B Partition Updates

MULTIPLE CHOICE

What is the purpose of an A/B atomic partition update in updaterd?

- **A.** To double the robot's walking speed.
- **B.** To safely test a new policy in Slot B with instant auto-rollback to Slot A if the robot trips.
- **C.** To install video games on the Rockchip CPU.

Q2: Automated Physics CI Gates

MULTIPLE CHOICE

Why must new neural policies pass 1,000 headless MuJoCo simulations in CI?

- **A.** To mathematically guarantee the model does not trip or spike motor torques before touching real hardware.
- **B.** Because physical robots do not have batteries.
- **C.** To make the simulation render faster in 3D.

Q3: Cryptographic Model Signatures

MULTIPLE CHOICE

How does cryptographic ED25519 signing protect an edge robot?

- **A.** It prevents the robot from getting wet in the rain.
- **B.** It ensures the robot only executes verified neural weights signed by authorized DevOps engineers, blocking rogue models.
- **C.** It encrypts the camera lens.

Q4: Fleet Observability & Metrics

MULTIPLE CHOICE

Why is real-time telemetry observability critical for a fleet of robots?

- **A.** It tracks motor temperatures, latency jitter, and fall rates to detect mechanical wear before catastrophic failure.
- **B.** To broadcast the robot's location to social media.
- **C.** It reduces the physical weight of the robot.

Q5: DevSecOps Release Pipeline

MULTIPLE CHOICE

In the 5-stage DevSecOps lifecycle, what happens after automated CI testing passes?

- **A.** The model is immediately deleted.
- **B.** The model is cryptographically signed and packaged for secure OTA deployment.
- **C.** The robot shuts down permanently.

Answer Key: 1-B, 2-A, 3-B, 4-A, 5-B (1-B: Safe A/B rollback; 2-A: Automated physics safety; 3-B: Cryptographic trust; 4-A: Predictive diagnostics; 5-B: Secure OTA release)